FEEL BEFORE YOU BUILD

Hands-on prototyping with actuators · 2 hours

AARHUS UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE

MULTIMODAL INTERACTION
PROTOTYPING WORKSHOP

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WHY WE ARE HERE

You can read a datasheet. You have never held one of these.

Most projects end up on the screen and the speaker

- Not because touch was wrong
- Because nobody knew a coin motor costs 12 kr

Today: decide it feels cheap, pick something else

- Better now than in week 46

HOW TODAY WORKS

Six benches, already wired and running

- Ten minutes each. Hard timer
- Change one parameter, answer one question
- You wire nothing today

Then build one signal

- Someone else has to read it

RUN SHEET

Times are offsets from the start. The rotation is the spine of the session, so keep it moving.

0:00	10 min	Two motors, same message. Which felt urgent?
0:10	60 min	Bench rotation, 10 minutes each
1:10	10 min	Round-the-room: what surprised you
1:20	30 min	Build one signal
1:50	10 min	Blind test and pack down

THE SIX BENCHES

01 ERM COIN MOTOR

VIBRATION

An off-centre weight on a motor shaft.

The same part as in your phone.

Speed and strength are welded together.

"Gentle but fast" is not on offer.

part	10mm 3V coin ERM
drive	2N2222 + 1N4148
pin	D9 (PWM)
draw	~75 mA
spin-up	20-40 ms
cost	~12 kr

CARD QUESTION

At what point does it stop feeling like information and start feeling like a malfunction?

02 LRA + PIEZO DISC

VIBRATION

Two ways out of bench 01's compromise.

LRA: resonance in 5 ms, stops just as fast.

The phone-keyboard feel.

Piezo: near-instant, almost no power.

A tick, not a thump.

```
parts    LRA 10mm + piezo
driver    DRV2605L (I2C)
pin    A4 SDA / A5 SCL
effects    123 waveforms
cost    ~90 kr
```

CARD QUESTION

Which of the three vibrators would you actually put on a wrist, and why

03 SOLENOID TAP

IMPACT

One 15 ms pulse against a fingertip.

The bench everyone remembers.

A knock reads as a person tapping you.

Buzzing reads as a machine.

part	5V push-pull
drive	MOSFET + flyback
pin	D6
peak	~1.1 A
duty	<= 25%, gets hot
cost	~45 kr

CARD QUESTION

How many taps before it goes from alert to nagging?

04 CAPACITIVE TOUCH

TOUCH INPUT

Bare finger on a surface. Servos react.

No button. No sensor you can point at.

The input is a wire behind a plate.

Foil, card, fabric. Anything works.

sense	ADCTouch on A0
baseline	25-sample roll
trigger	1.01x average
debounce	100 ms
servos	D4 + D13
cost	~0 kr, a wire

CARD QUESTION

Put your hand near the plate without touching. When exactly did it decide that was a touch?

BENCH 04: THE WHOLE TRICK

Touch input for the price of a wire.

reading > baselineAverage() * 1.01

Threshold is relative, never absolute

• Readings drift with humidity and mains hum

• Fixed threshold works at 9am, fails at 1pm

Baseline only learns while untouched

• Or a long press teaches it that a finger is normal

05 PELTIER WARM / COOL

THERMAL

Heats one side, cools the other.

Flip the current, it reverses.

Sit with it. Seconds before you are sure.

Hot alternating with cold reads as lukewarm.

```
part    TEC1-12706
drive    L298N H-bridge
pin    D3 PWM / D4-5
draw    2-4 A, bench PSU
onset    3-8 s
cap    45 C in software
```

CARD QUESTION

Time yourself. How long until you would bet money on warmer vs cooler?

06 THE SAME SIGNAL, TWICE

AUDIO + TOUCH

One driver. 40 Hz you feel, 400 Hz you hear.

Toggle them alone, then together.

Together is not louder. It is more certain.

Two channels, same message. Cheap reliability.

part	bone-conduction
amp	PAM8403 class-D
pin	D11 (tone)
felt	~40 Hz
heard	~400 Hz
cost	~110 kr

CARD QUESTION

With ear defenders on, does the signal still work?

INPUTS, AND PRIOR ART

INPUTS WORTH KNOWING ABOUT

One table, not a rotation. Wander over during the build.

- Capacitive touch. A wire. See bench 04
- Heart rate (PPG). Useless once the hand moves
- Skin conductance (GSR). Relative change only
- Flex sensor. The cheap data glove
- Pressure (FSR). Calibrate every pad
- Your own phone. No soldering at all

TWO RIGS THAT ALREADY EXIST

Built here, by students at roughly your stage.

Moving screen, Arduino Uno

- Breathes when idle. Retreats when touched
- Easing and idle motion do the work. A dozen lines

Instrumented sock, ESP32

- Six force sensors under a foot, batched over Wi-Fi
- Calibrate per sensor. Batch your writes

BUILD ONE SIGNAL

THE BRIEF

30 minutes. Not long enough to be precious about it.

One actuator. Three messages

- Arrived. Something wrong. Finished

Vary two things only

- Rhythm of the pulses
- How strong they are

Blind test: they name all three

- Note which two got confused, and why

THE CONSTRAINT THAT MATTERS

The recipient cannot look at the device.

If it only works while someone watches a screen,

you built a visual interface with a motor glued to it.

GO AND TOUCH THINGS

github.com/gust1527/multimodal-actuator-workshop

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